

Neglect and Related Disorders

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ABSTRACT

Neglect is a failure to report, respond, or orient to contralateral stimuli that is not caused by an elemental sensorimotor deficit. Subtypes of neglect are distinguished by input (attentional) or output (intentional) demands, the distribution (personal, spatial, and representational), and the means of eliciting the signs (unilateral or bilateral stimuli). In this article we discuss how to assess patients for neglect, the pathophysiology of neglect, and the treatment of neglect.

Keywords: Neglect, unilateral neglect, hemispatial neglect, inattention

Neglect is a common and severely disabling neurobehavioral disorder induced by discrete lesions such as cerebral infarctions and hemorrhages, tumors, trauma, and degenerative diseases. Studies have shown that patients with neglect are less likely to be able to live independently than even patients with severe aphasia and a right hemiparesis. In this article, we will first discuss the signs and symptoms associated with this disorder and the means by which patients may be assessed for neglect and related disorders. We will then discuss the patho-

physiology of neglect. In the last section we will discuss the management and treatment of neglect.

CLINICAL SIGNS AND SYMPTOMS

DEFINITION

Neglect is the failure to report, respond, or orient to meaningful or novel stimuli.¹ This failure is primarily

Objectives

On completion of this article the reader will be able to recognize and distinguish neglect, inattention, and subtypes of neglect and be better able to test for these phenomena in the clinical setting.

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for stimuli or actions that occur on the side contralateral to a hemispheric lesion. If this failure can be accounted for by either an elemental sensory or motor defect, then the failure is not considered to be neglect.

SUBTYPES

Many subtypes of neglect have been reported. The subtypes are distinguished by several different factors including (1) input or output demands (inattention or sensory neglect versus intentional or motor neglect), (2) means of eliciting (unilateral versus bilateral stimuli or movements), and (3) distribution: personal neglect versus spatial neglect.

SENSORY NEGLECT

Sensory neglect or inattention is a form of selective unawareness. Sensory neglect may be defined by its modality (e.g., tactile, visual, auditory) and its distribution (e.g., hemispatial or personal). Sensory neglect may also be associated with an ipsilesional attentional bias and an inability to disengage from stimuli in ipsilesional space.

To test for inattention, the patient is presented with stimuli (visual, tactile, and auditory modalities) or no stimuli to either the ipsilesional or contralesional side in random order. Each time a stimulus or nonstimulus is presented, the examiner says "now." If patients make more contralesional than ipsilesional errors, they have evidence of inattention.

If a patient consistently fails to detect a contralesional stimulus, it may be difficult to learn whether the problem is inattention or an elemental sensory defect. There are several means by which these deficits can be differentiated. In the auditory modality, because hemispheric lesions do not cause contralesional deafness, if a patient fails to detect contralesional auditory stimuli, the patient has inattention. In the visual and tactile modality, inattention may be hemispatial rather than hemiretinal or limited to one side of the body. Therefore, patients who cannot detect contralesional stimuli may be able to detect these stimuli if their eyes are deviated to the ipsilesional side (such that the contralesional retinal field now falls in ipsilesional hemispace). Caloric stimulation and psychophysiological techniques such as evoked potentials or galvanic skin responses may also help demonstrate that a defect is inattention rather than deafferentation.

Many patients who are first unaware of stimuli may recover and be able to detect isolated stimuli. However, when presented with two or more simultaneous stimuli, they may demonstrate unawareness of the contralesional stimulus (extinction).

INTENTIONAL OR MOTOR NEGLECT

There are several forms of intentional neglect. For example, patients who appear to be hemiparetic may not have damage to the corticospinal system as determined by imaging or magnetic stimulation.² This pseudohemi-

paresis is termed limb akinesia or motor neglect. Other subjects may be unable to move a limb in contralesional hemispace but may be able to move it in ipsilesional hemispace.³ This disorder has been termed hemispatial akinesia. Similarly, patients may be impaired in moving the head, eyes (e.g., gaze paresis), or even the arm in a contralesional direction (directional akinesia).⁴ Some patients who can move the contralesional limb alone have difficulty initiating movement of this limb at the same time as they are starting to move the ipsilesional limb (motor extinction).⁵

Hypokinesia is a milder form of intentional neglect in which there is a delay in initiating a movement. One of the best means of testing for hypokinesia is by performing reaction times. These can be performed for the limbs (limb hypokinesia), head, or eyes in a hemispace (hemispatial akinesia) and in a direction (directional hypokinesia).

Movements of reduced amplitude (hypometria) may also occur in the limbs, eyes, or head. This may also be directional or hemispatial. To test for hypometria, one directs the patient to move a certain distance and then ascertains whether the subject undershot his or her mark. However, one must use a testing procedure that enables one to dissociate inattention (to the target) from hypometria. For example, before having a patient saccade to a target, the patient should be able to see the target. If the patient then makes multiple small saccades until he or she reaches the target, the patient has a directional hypometria of the eyes.

Motor impersistence is the inability to sustain a movement or posture. In general, in testing for impersistence, the examiner asks the patient to maintain a posture for a given period of time. Motor impersistence may also be seen in the limbs, eyes, or head. It may also be hemispatial or directional. One usually asks a patient to maintain a posture for 15 seconds (e.g., "look leftward and keep looking leftward until I tell you to stop").

HEMISPATIAL NEGLECT

Spatial neglect may occur in all three dimensions of space: horizontal (right, left), vertical (up, down), and radial (near, far). Therefore, tests that assess spatial neglect should test all three dimensions.⁶ There are three bedside tests used to test for spatial neglect: line bisection, target cancellation, and drawing. Each of these tests may assess different functions. When performing the line bisection task, a patient is presented with a line and asked to find the middle of the line. Longer lines are more likely to detect neglect than are shorter lines. With very short lines, a patient with neglect may err by placing their mark in the portion of the line that is in contralesional hemispace.⁷ Placing the line in contralesional hemispace is more likely to detect neglect than placing the line in ipsilateral hemispace.⁸ Putting cues on the ipsilateral side of the line may also increase neglect, and having cues on the contralesional side may reduce new neglect.

When performing the cancellation test, a sheet of paper with targets is placed before the patient, and the

patient is asked to mark out or cancel all the targets. Increasing the number of targets, placing them randomly on the sheet, and using foils that are difficult to discriminate from the targets are methods that can be used to increase the sensitivity of the task. It is important to see the means by which the patient explores the sheet (i.e., from left to right versus right to left) while the patient is performing the cancellation task. Most normal people explore from left to right. Patients with left-sided neglect often start from the right and explore right to left.

Drawing should be tested both by having the patient copy the examiner's drawing and by having the patient draw spontaneously. Copying asymmetric nonsense figures is usually a more sensitive test for neglect than is copying symmetrical familiar figures. The more detailed the drawing, the more sensitive the task. A patient who has difficulty copying one side of a figure may have an attentional or an intentional deficit. A subject who does not draw one side spontaneously may have an intentional or representational deficit (i.e., the memory of one side of the objects is no longer present because it was destroyed by the lesion).

To dissociate attentional from intentional hemispatial neglect, one can use different forms of the crossed response task in which the subject acts in ipsilateral space when presented contralesional stimuli and acts in contralesional space when presented with ipsilesional stimuli.^{9,10} This can be performed using a variety of techniques including video cameras with monitors, strings and pulleys, and mirrors. To dissociate representational from intentional deficits, one can use the fixed aperture technique in which an opaque screen covers the sheet that contains the targets and has an aperture that allows the subject to see only one target at a time. In one half of the trials, the patient moves the top sheet, and in the other half the subject moves the lower sheet. Whereas a failure to explore one part of the sheet in both conditions suggests a representational deficit, a failure to explore opposite sides¹¹ of the sheet in the two different conditions suggests a motor intentional deficit.

When patients have hemispatial neglect, they can be neglecting the contralesional part of space that is opposite their eyes, head, or body (body-centered frame of reference).¹² Alternatively, they can be using an environmental frame of reference. One can dissociate body-centered from environmentally centered neglect by having the patients bisect lines in the upright position and then having them bisect lines when lying down on their right or left sides. If their performance does not change with a change of positions, their neglect is viewer centered; if their neglect changes, it is environmentally centered.

PERSONAL NEGLECT

Patients with personal neglect may fail to groom or even dress one side of their body. They may complain that someone is in bed with them. To test for personal neglect more formally, one can ask the subjects to show or point to different portions of the right or left side of their body. One can perform a cancellation type of task in which one puts little pieces of sticky paper on differ-

ent portions of the body and asks the subject to take off all the pieces of paper. Lastly, one can perform a bisection task in which a horizontal line is placed before the subject, who is asked to show, by pointing, the position on the line that is directly across from the middle of his or her chest (e.g., the midsagittal plane).

REPRESENTATIONAL IMAGERY NEGLECT

Representational imagery neglect may be spatial¹³ or personal.¹⁴ For example, when patients are asked to recall details of a familiar scene, they may report primarily the details that are ipsilateral to their lesion.¹³ Similarly, they may also fail when performing a task in which they are required to imagine a contralesional part of their body.¹⁴

PATHOPHYSIOLOGY OF NEGLECT

ATTENTION

Patients with neglect and related disorders often appear to be unaware of contralesional stimuli, and because this unawareness cannot be accounted for by deafferentation, it is thought that many symptoms of neglect are related to attentional deficits.

Attention is a construct that is difficult to define. Perhaps this is so because attention is a mental process rather than an object. Humans have a limited processing capacity and receive more afferent stimuli than they can possibly process. In addition to the external stimuli, humans can activate internal representations of stimuli and these internal stimuli can also be further processed (thinking). The processing of these internal stimuli may further tax a limited capacity system and reduce a person's ability to process afferent stimuli. Because humans have a limited processing capacity, they need a means to triage incoming information. Attention is the mental process that permits humans to apply triage to afferent input. Triage status depends on the potential importance or significance of the incoming information to the organism. One cannot know the potential importance of novel stimuli and therefore one must, at least temporarily, attend to novel stimuli until their significance is determined. Significance is determined by two major factors: goals or sets and biological drives or needs. A stimulus that is important for that person's goals, sets, needs, or drives will be assigned a higher level than a stimulus that is irrelevant.

After determining stimulus significance, the organism must direct attention to the significant stimulus and away from the irrelevant stimulus. There are at least two means by which one may be able to direct attention. One can use sensory modalities. For example, when reading this article, which I hope is important for your long-term goals, you are directing your attention to visual input and are not aware of how your left foot feels in your shoe until I mention it. However, because the somatosensory information from your left foot is irrelevant after a few minutes, you will no longer be aware of the sensation

coming from this foot. Another means for directing attention is spatial. Therefore, if this journal is on your lap and you are sitting in a chair, you will direct attention to the lower portion of body-centered space and be unaware of stimuli in the upper portion of visual space.

It is not completely known how the brain mediates attention. In general, there are three models: the efferent gaiting model of Hernandez-Peon et al,¹⁵ the cortical-reticular model of Sokolov,¹⁶ and the mixed model of Watson, Valenstein, and Heilman.¹⁷ We will briefly give an overview of each of these models.

Hernandez-Peon, in a series of influential papers, proposed that the central nervous system sent efferent fibers to peripheral portions of sensory systems and that these efferent projections regulated the amount of afferent input into the central nervous system. To test this hypothesis, he and his coworkers recorded evoked potentials from the sensory (cochlear) nucleus of the cat's auditory system in response to clicks when the cat was not looking at any significant stimulus that attracted its attention. However, when they put a mouse in the cage with the cat and the cat paid visual attention to this mouse, the auditory evoked potential in the sensory ganglia was reduced. There are efferent projections to the peripheral sensory systems from the central nervous system, and there may be some efferent gating. However, if efferent gating was the major means by which we pay attention, it would reduce our chances of survival. For example; while reading this article and paying visual attention, if we gated somatosensory input peripherally and a spider crawled into our left shoe, we would be unable to divert our attention and we would be unaware of the spider. Fortunately, this is not the case; if a spider was crawling around your foot, no matter how interesting you found this article you would divert your attention to your foot. One of the problems with the studies of Hernandez-Peon and his coworkers is that they used speakers rather than earphones, and cats have ears that move toward the stimulus to which they are attending (i.e., toward the mouse and away from the speakers). When this study was repeated using earphones, Hernandez-Peon's findings could not be replicated.

In contrast to Hernandez-Peon, Sokolov thought that a cortical-reticular system loop was responsible for mediating attention. Since the work of Moruzzi and Magoun,¹⁸ it has been known that the reticular activating system (RAS) is important in mediating arousal. When an organism is presented with a novel or significant stimulus, it becomes behaviorally and electrophysiologically aroused. An organism that cannot become aroused cannot normally attend. Therefore, arousal is a critical element in the systems that mediate attention. However, attention must also be selective, and Sokolov posited that it is the cortex that selects which stimuli to attend and which stimuli not to attend. According to Sokolov, incoming stimuli are processed by the cortex. The cortex makes a model of these incoming stimuli, and if the stimulus is novel, the cortex activates the RAS. If the stimulus matches the model and is therefore not novel, the RAS is not activated (habituation).

Whereas Sokolov's model may account for why we are unaware of our foot (no novel stimulus) until the spi-

der crawls on it, this model cannot explain spatially selective attention and "top down" attention (e.g., attending to a stimulus because of motivational states and long-term goals or set). Sokolov's model also did not specify the portions of the cortex that perform these selective function in humans.

The model we have proposed¹ not only specifies the anatomic regions of the cortex that compute significance and spatially directs attention but also attempts to account for "top down" or motivationally directed attention. In addition, it attempts to account for habituation using a thalamic gating postulate.

Until recently (with the advent of functional imaging), almost all that we have learned about how the human brain mediates attention came from studying patients who had discrete lesions of their brain. In the early part of the 20th century, investigators such as Poppelreuter suggested that this disorder, termed hemispatial or unilateral neglect, may be related to a deficit of awareness or an attentional disorder. However, because Bender¹⁹ could not alter neglect by asking subjects to attend to their contralesional side, he thought that the disorder was a sensory-perceptual deficit. Denny-Brown and Banker²⁰ called the disorder amorphosynthesis because they thought that there was a defect in sensory synthesis that led to a perceptual and representational deficit. Denny-Brown and Banker²⁰ and Bisiach and his coworkers¹⁰ demonstrated that patients with neglect can also fail to report the contralesional portion of visual images (representational neglect). In addition, patients with neglect may also fail to recall stimuli that were presented to their contralesional side in spite of originally being aware of these stimuli. Although there is evidence that some patients have a representational deficit such that they have no concept that the space contralateral to their hemisphere lesion exists, there is also evidence that some of the signs and symptoms of hemispatial neglect are being induced by attentional defects. For example, when some patients with left-sided neglect are given cues that are either "bottom up," such as a flashing light on the contralesional side, or "top down," such as instructing a patient to read a letter on the left, neglected side, before searching, their performance on tests for spatial neglect, such as line bisection, improves. In another test for spatial neglect, the cancellation test, one can increase attentional demands by increasing the number of stimuli in the array or increasing the number of distracting stimuli. That patients with neglect perform worse under these more difficult conditions provides further support for the postulate that these patients have a reduction of their attentional capacity. Lastly, patients with neglect perform more poorly on tasks that require focused attention (e.g., distinguishing targets from distractors) than they do on tasks that have less attentional demands.

Because hemispatial or unilateral neglect in some patients appears to be an attentional disorder, studying patients and experimental animals with this disorder has led to the development of a neuropsychological model of how the brain mediates attention.

In humans, neglect is most commonly associated with lesions that involve the right-sided inferior parietal

lobe.²¹ However, there are other areas where lesions in humans have been reported to induce neglect. These include the dorsolateral frontal lobes,²² the mesial frontal lobes including the cingulate gyrus,²² the thalamic and mesencephalic reticular formation,¹⁷ and the basal ganglia.²³

Because there are limits on the types of research that can be done on humans, much of what we have learned about neglect comes from research on Old World monkeys. Unlike that of humans, monkeys' inferior parietal lobe (IPL) is Brodmann's area 7. In humans, the intraparietal sulcus separates the parietal lobe into the IPL (areas 40 and 39) and the superior parietal lobe (SPL), Brodmann's areas 7 and 5. Whereas some have thought that monkeys' IPL is the homologue of the human IPL, others believe that both banks of monkeys' superior temporal sulcus (STS) are the homologue of the human IPL. Watson et al²⁴ demonstrated in Old World monkeys that neglect (unawareness) is associated with STS rather than IPL lesions, thereby supporting the postulate that monkeys' STS is the homologue of humans' IPL.

Monkeys' STS is composed of multiple subareas and is one of the sites of multimodal sensory convergence. Not only does the auditory, visual, and somatosensory association areas project to STS, but STS also receives projections from other convergence areas such as monkeys' area 7. Because ablation of sensory convergent areas such as monkeys' area 7 does not produce neglect, sensory convergence alone cannot account for neglect.

Haxby et al²⁵ suggested that whereas the ventral portion of the visual system (e.g., the ventral temporal lobes) is important for determining the type of stimuli (What?), the dorsal portion (e.g., Brodmann's area 7) codes the spatial location (Where?). The somatosensory and auditory systems may also be divided into "what" and "where" subsystems. Watson and coworkers proposed that these "what" and "where" systems converge and integrate in monkeys' STS or humans' IPL. According to Watson and coworkers, STS lesions in monkeys or IPL lesions in humans may cause neglect or unawareness because this area not only receives polymodal input but also is the convergence site of the perceptual and cognitive systems that deal with the what and where of environmental awareness. The STS also receives input from both the cingulate gyrus and the dorsolateral frontal lobes. The dorsolateral frontal lobes are important in mediating goal-oriented behaviors and may provide the STS information about long-term goals. The cingulate gyrus is part of the limbic system and may provide the STS information about biological needs and drives. Because monkeys' STS or humans' IPL receives input from the perceptual/cognitive systems and receives conative and motivational information, it may be an ideal place to make attentional computations.

The STS not only has reciprocal relations with the cortical areas from which it receives information but also appears able to modulate the activity of the reticular activating system (cortical control of arousal). Therefore, lesions of monkeys' STS or humans' IPL impair the organisms' ability to make attentional computations and to arouse or activate the areas that determine "what" and "where."

The STS also has strong reciprocal relations with the hippocampus. Damasio has posited that the hippocampus is important for the retroactivation of sensory association areas. Therefore, lesions of monkeys' STS or humans' IPL may interfere with retroactivation and induce the imagery and representational defects described by Denny-Brown and Bisiach.

The corticallimbic reticular model we have discussed cannot fully account for habituation (not feeling that foot). Sensory information that reaches the cortex is relayed through sensory-specific thalamic nuclei. The reticular nucleus of the thalamus (NR) is a thin nucleus that envelops the thalamus and sends inhibitory fibers into the sensory relay nuclei of the thalamus. When the NR is activated, it inhibits thalamic relay to the cortex. The NR receives projections from the cortex that may inhibit this inhibitory nucleus. Therefore, when a stimulus is neither novel nor significant, corticofugal projections to the NR may decrease thalamic transmission to the cortex and vice versa. Support for this NR hypothesis comes from evoked potential and imaging studies demonstrating that attending to somesthetic stimuli activates the primary cortex.¹⁷

INTENTION

Some patients with hemispatial neglect are aware of contralesional stimuli, but fail to act on these stimuli. Although not weak, these patients fail to act on contralesional stimuli because they have a defect of intention.

As we discussed, attention is the process by which the brain applies triage to incoming stimuli according to their significance. However, the brain not only has a limited capacity to process incoming information, it also has a limited ability to prepare for action and to act. The process by which the brain applies triage to actions we call intention. Failure of this intentional system can cause an inability to initiate movements termed akinesia. Milder defects of intentional systems may induce a delay in the initiation of movement called hypokinesia. In some cases movements must be sustained and a failure to sustain movements is called impersistence.

It had been reported that dorsolateral frontal lobe lesions (Brodmann's area 8) in monkeys induce sensory neglect.²⁶ Because of its connectivity, we thought that the dorsolateral frontal lobe may play a more important role in intention than in attention. To learn whether the frontal lobes play an important role in mediating intention, we trained monkeys in a crossed response paradigm such that when stimulated on the left the monkey responded with its right arm and when stimulated on the right the monkey responded with its left arm. After unilateral dorsolateral frontal lobe lesions, the monkeys responded with their ipsilesional arm to contralesional stimuli but failed to respond to ipsilesional stimuli with their contralesional arm. Because the motor areas were not damaged and these animals were not weak, these results suggest that the frontal lobes play a critical role in the intentional systems.²⁷

The intentional and attentional systems have to be interactive, especially for tasks such as exploration. Therefore, not only do the frontal lobes have to influence the parietal lobes but the parietal lobes must influence the frontal lobes. The STS in monkeys has strong reciprocal projections with the arcuate and periacuate regions of the dorsal lateral frontal lobe. The arcuate region or frontal eye field is important for the initiation of purposeful saccades, and the periacuate region is important for the initiation of voluntary arm movements. It has been demonstrated that lesions of this region as well as this region's projections to the basal ganglia, cingulate gyrus, and thalamus are part of a motor-intentional system that, when injured, may be associated with motor-intentional neglect.¹ Lesions of monkeys' STS or humans' IPL may be associated with exploratory failures and elements of motor-intentional neglect because of the critical input the dorsal lateral frontal lobe receives from the STS.

RIGHT-LEFT ASYMMETRIES OF ATTENTION AND INTENTION

Neglect can be seen with either right or left hemisphere lesions. However, in humans, neglect is more common and severe with right hemisphere lesions than it is with left hemisphere lesions.²⁸ These asymmetries of neglect appear to be related to asymmetrical attentional representations of the body and space. Electrophysiological and functional imaging studies suggest that whereas the left hemisphere attends primarily to the right side or in a rightward direction, the right hemisphere can attend to both sides or in both directions.^{29,30} In a similar fashion, although the left hemisphere prepares the right side for action (or prepares for rightward actions), the right hemisphere can prepare both sides for action (or prepare for action in either direction).³¹ Lastly, the right hemisphere appears to have more control of the reticular activating system than does the left. For example, right parietal lesions are associated with a greater reduction of an arousal response than are left hemisphere lesions.³²

TREATMENT OF NEGLECT

TREAT UNDERLYING DISEASES

The behavioral manifestations of the neglect syndrome are induced by underlying cerebral disease or dysfunction. Therefore, the clinician should primarily evaluate and treat the underlying disease and, when possible, prevent further brain injury.

MANAGEMENT OF THE ENVIRONMENT

Attentional and inattention disorders dramatically reduce an organism's capacity to interact with the environment; therefore, any patients with these disorders must have their environment designed to reduce the risk of injury and to increase the possibility that the patient could successfully interact with environmental stimuli.

For example, patients with even the mildest attentional disorders should not drive or use machinery that has the potential for injury to themselves or others. Neglect is often in reference to the person's midsagittal plane rather than to the object in the environment. Therefore, the patient should be positioned and the environment arranged such that environmental stimuli take place on the right side of the patient's body-centered space. Because patients with neglect have a reduced attentional capacity and have difficulty dealing with more than one stimulus, there should be a reduction of competing or distracting stimuli.

Because it is possible that stimuli presented to the contralesional fields and body parts may be therapeutic, when detecting stimuli is not critical for their well being, patients should receive stimuli in their neglected fields.

PRISMS

When testing neglect, one often finds that moving the actual objects into ipsilesional hemispace may improve detection. Prisms may move visual images, and the use of prisms with adaptation has been used to treat neglect.³³

SHIFTING ENVIRONMENT

There are other means of shifting (e.g., left to right) a subject's perception of visual space. Cold caloric stimulation of the left ear,³⁴ optokinetic nystagmus,³⁵ and vibration of the neck muscles³⁶ can also improve neglect, but there is little evidence that these effects last for many minutes after the stimulation has terminated.

TRAINING

Several investigators have demonstrated that by using operant techniques, patients with neglect can be trained to explore and scan contralesional hemispace.³⁷ However, this training often remains task specific and often fails to generalize to other tasks.

CUEING

Many symptoms of the neglect syndrome are related to an inability to move attention to the contralesional hemispace or a bias to attend to ipsilateral hemispace. One means of getting the patient to attend to contralesional hemispace is to provide attentional cues. Novel stimuli are powerful cues (bottom up cues). Bottom up visual,³⁸ auditory,³⁹ and tactile cues have been used to help patients with neglect. Directions to attend to a side (top down) may also help neglect. However, if the goal of rehabilitation is to have patients live independently, this cueing strategy has to be "internalized," and this is what Diller and Weinberg³⁷ attempted to do in their training paradigm (e.g., search for a cue on the left before acting on the environment).

Many symptoms of neglect are caused by motor-intentional deficits such that subjects fail to act in or toward contralesional hemispace. In addition, preparing to act in a portion of space may also help one direct attention to the same part of space. When performing most tasking, right-handers prefer to use their right hand, and this hand preference may even grow stronger after a right hemisphere stroke that may be associated with a left hemiplegia or limb akinesia or both. However, Halligan and Marshall⁴⁰ demonstrated that when a patient with neglect from a right hemisphere lesion performs with the left hand, the performance is more normal than when the patient performs with the right hand. Although many patients with neglect have hemiplegia, Robertson and North⁴¹ showed that any attempted movement of the left arm improves performance. Finger movement, even when the left arm could not be viewed by the patient, appeared to improve performance. Some subjects used their left arm as a cue. When subjects were trained to place their left arm to the left margin of any activity, their performance improved.

EYE PATCHING

Sprague⁴² induced neglect (asymmetrical exploration and orienting) in animals by making a posterior neocortical lesion. He subsequently reduced this asymmetrical orienting and exploration by ablating the superior colliculus on the opposite side. Each eye projects more strongly to the contralateral colliculus than it projects to the ipsilateral colliculus. Posner and Rafal⁴³ suggested that patching one eye (e.g., the right) may help patients with neglect (e.g., left) because patching one eye may reduce the activation of the opposite colliculus. Butter et al³⁸ tested this means of treatment and did demonstrate a reduction of left-sided neglect with a right eye patch. Barrett et al⁴⁴ found, however, that one subject with left-sided neglect was made worse by patching the right eye.

PHARMACOLOGICAL TREATMENT

In experimental animals, neglect can be induced by unilaterally interrupting the dopaminergic system. Corwin et al⁴⁵ demonstrated in rats that dopamine agonists reduce neglect. Fleet et al⁴⁶ reported two patients who improved on tests of neglect when treated with the dopamine agonist bromocriptine. However, double-blinded controlled studies have not been performed using this treatment. Some neurologists have indicated that some of their patients with neglect who were treated with bromocriptine did not improve, and it is even possible that some patients may get worse. For example, Barrett et al⁴⁴ treated a patient with dopamine agonist therapy, and the patient's performance on tests for spatial neglect became worse during treatment. The patient, however, had a lesion that involved subcortical structures including the putamen. Therefore, if one elects to treat a patient with dopamine agonists, it is recommended that the patient be carefully tested before treatment, during treatment, and after at least one withdrawal period.

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